

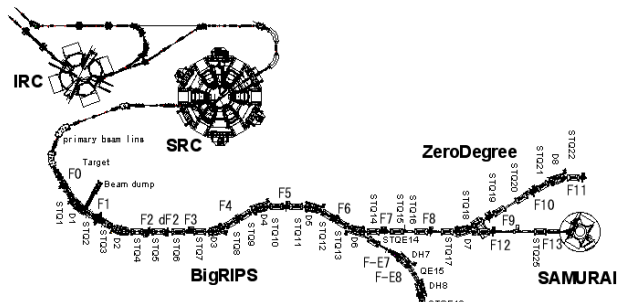
Why SAMURAI73 Needs Two Polarimeters

Full tensor reconstruction and locating p_{yy} loss

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IVR needs the tensor state delivered to the target



First IVR run

Start with vertical p_{yy} ;
longitudinal p_{zz} is harder
to prepare.

After SRC

What polarization leaves
the accelerator chain?

After BigRIPS

What polarization reaches
SAMURAI73?

Spin-1: 8 general degrees, 5 tensor, 1 prepared

For $w_a \geq 0$ and $\sum_a w_a = 1$,

$$\rho = \sum_a w_a |\psi_a\rangle \langle \psi_a|.$$

A normalized 3×3 Hermitian ρ has

$$8 = 3_{\text{vector}} + 5_{\text{tensor}}$$

independent real parameters.

An axial tensor prepared about $\hat{n}$ is

$$\mathbf{P}_0^{(2)}(q, \hat{n}) = \frac{q}{2} (3\hat{n}\hat{n}^T - I).$$

$$\hat{n} = \hat{y} : \quad \mathbf{P}_0^{(2)} = \frac{p_{yy}}{2} \text{diag}(-1, 2, -1),$$

$$\hat{n} = \hat{z} : \quad \mathbf{P}_0^{(2)} = \frac{p_{zz}}{2} \text{diag}(-1, -1, 2).$$

Tensor only: 5 degrees. Fixed preparation axis: one amplitude q .

Precession is measured in the local beam frame

1. Magnetic field rotates the spin in the lab

$$\rho_{\text{lab},f} = U_s \rho_0 U_s^\dagger, \quad R_s \in SO(3).$$

Thomas–BMT determines R_s along the trajectory.

$$R_{\text{rel}} = R_o^T R_s,$$

2. The downstream beam defines new axes

$$\rho_{b,f} = U_o^\dagger \rho_{\text{lab},f} U_o.$$

R_o follows the transported momentum and defines the local x_b, y_b, z_b frame.

$$\mathbf{P}_{b,f}^{(2)} = R_{\text{rel}} \mathbf{P}_0^{(2)} R_{\text{rel}}^T.$$

Lowest-order model: one common R_{rel} moves the polarization axis but preserves the tensor eigenvalues and magnitude.

Different trajectories can reduce the ensemble polarization

Beyond the central ray, each phase-space point $\xi = (x, x', y, y', \Delta p/p, \dots)$ has its own calculated rotation:

$$\mathbf{P}_f^{(2)}(\xi) = R_{\text{rel}}(\xi) \mathbf{P}_0^{(2)} R_{\text{rel}}^T(\xi), \quad \overline{\mathbf{P}^{(2)}}_f = \int d\xi f(\xi) \mathbf{P}_f^{(2)}(\xi).$$

1 degree

fixed axis and one
amplitude q

3 degrees

one rigid tensor: q plus two
axis angles

5 degrees

a general beam-averaged
tensor

Each particle remains coherent. The average can lose tensor magnitude because the particles arrive with different axes.

One station sees 4/5 tensor degrees; two see 5/5

Nominal p_{yy} monitoring

$$\frac{N_L + N_R - N_U - N_D}{N_L + N_R + N_U + N_D} \implies p_{yy}.$$

One station measures the amplitude if the beam is assumed to remain a pure p_{yy} state.

General-state test

One L/R/U/D station is blind to tensor p_{xy} and vector p_z .

A complementary second view rotates these directions into measured combinations.

Calibrated multi-angle measurement	tensor state	full density matrix
one station	4/5	6/8
two stations with characterized complementary transport	5/5	8/8

Two stations locate a true loss of polarization

Use an axis-independent tensor magnitude

$$Q \equiv \sqrt{\frac{2}{3} \text{Tr}[(\mathbf{P}^{(2)})^2]}, \quad Q = |p_{yy}| \text{ for an ideal axial } p_{yy} \text{ state.}$$

What happened?

$Q_2 \simeq Q_1$, but the principal axis changes:
coherent rotation, not depolarization.

$Q_2 < Q_1$ after transport correction: true
loss of beam-averaged polarization.

Where did the loss occur?

Q_1 already low after SRC: source,
acceleration, or SRC.

Q_1 nominal, but Q_2 low after BigRIPS:
between the stations, including BigRIPS
transport or acceptance.

A lower p_{yy} component alone is ambiguous: its axis may have moved, or its magnitude may have fallen.

Who is the contact for an after-SRC polarimeter?

Could a polarimeter be installed just downstream of SRC?

Who should we contact to discuss

- ▶ the available beam-line location and longitudinal space;
- ▶ vacuum, mechanical, electrical, and detector interfaces;
- ▶ beam optics, operation, radiation safety, and installation constraints?

Requested next step: identify the responsible beam-line contact and arrange an interface discussion.